MIL-S-21521A (Wep)
AMENDMENT 2
1 December 1964
Superseding
Amendment-1
9 November 1964

MILITARY SPECIFICATION

SPRINGS, LIQUID

This amendment forms a part of Military Specification MIL-S-21521A, 24 August 1964, and has been concurred in by the Bureau of Naval Weapons.

Page 6, paragraph 3.2.12: Delete and substitute:

"3.2.12 Charging Pressure. The internal charging pressure of the liquid springs, using the fluid specified in 3.2.11, shall be 13,500 ± 100 psig for the horizontal liquid spring and 11,800 ± 100 psig for each type of vertical liquid spring when gaged at a temperature of +80° ± 3°F."

Page 10, paragraph 3.3.2: Delete and substitute:

"3.3.2 Leakage With No Deflection (Accelerated). (See 6.1) When charged as specified in 3.2.12 (± 600 psig) with silicone fluid having a viscosity of 2 centistokes, the liquid spring shall show no loss in pressure in excess of 375 psi when allowed to stand under no deflection for 3 days. During that period, the spring temperature shall be not less than +70° nor more than +83°F. and shall be maintained at a constant temperature within a total range of 4°F. Thereafter, when charged with silicone fluid having a viscosity of 50 centistokes, the liquid spring shall be capable of meeting all performance requirements of this specification. The pressure loss shall be calculated by the following formula:

$$P = P_1 - P_2 + C\,(t_2 - t_1)$$

Where: P = Pressure loss - (psi)

P_1 = Initial pressure - (psig)

P_2 = Final pressure - (psig)

C = Temperature coefficient: 70 psi per degree Fahrenheit

t_1 = Initial temperature of spring degrees Fahrenheit

t_2 = Final temperature of spring degrees Fahrenheit

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Page 11, paragraph 3.3.4: Delete last sentence and substitute:

"Thereafter, the liquid spring shall show no damage and shall be capable of meeting all performance requirements of this specification."

Page 13, paragraph (c) of TABLE IV: Delete "20.6 ± 0.5 inches" and substitute "20.7 + 0.5 - 0 inches."

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